

SKIM: Efficient Video QA via Reinforcement Learning

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Background on Video QA

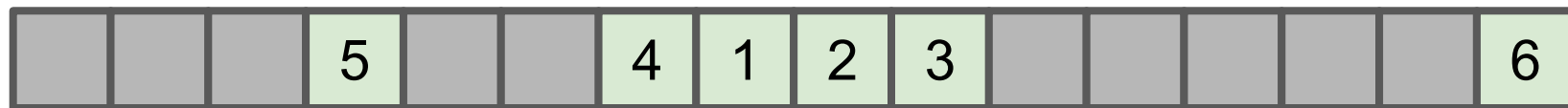
Passing all frames = inefficient →



- **Method 1:** Pass frames uniformly (lower frame-rate)



- **Method 2:** Text-frame matching → K frames with embeddings closest to query



- **Method 3:** Model loops through all frames, selects variable-sized subset

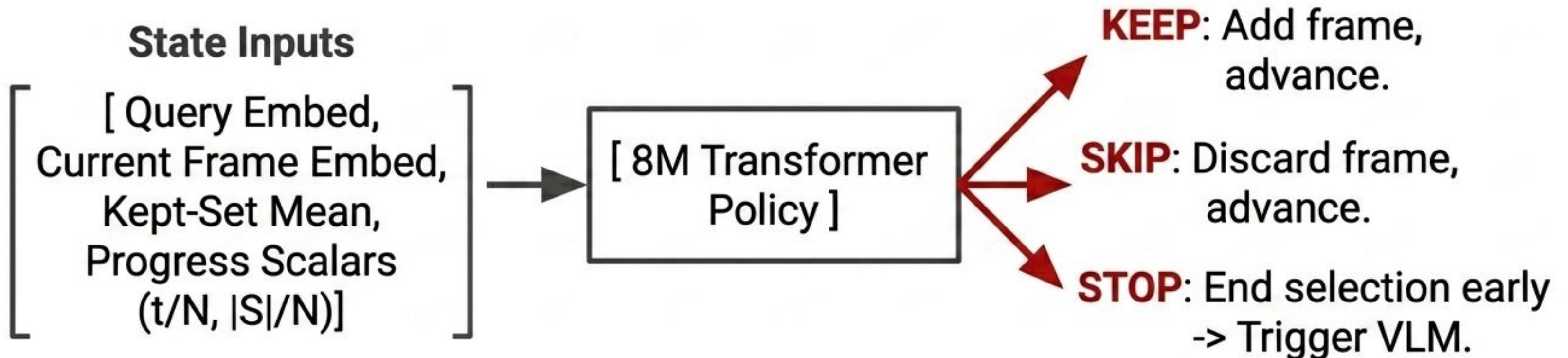


- **Our Goal:** Lightweight model to loop through frames sequentially, potentially stopping



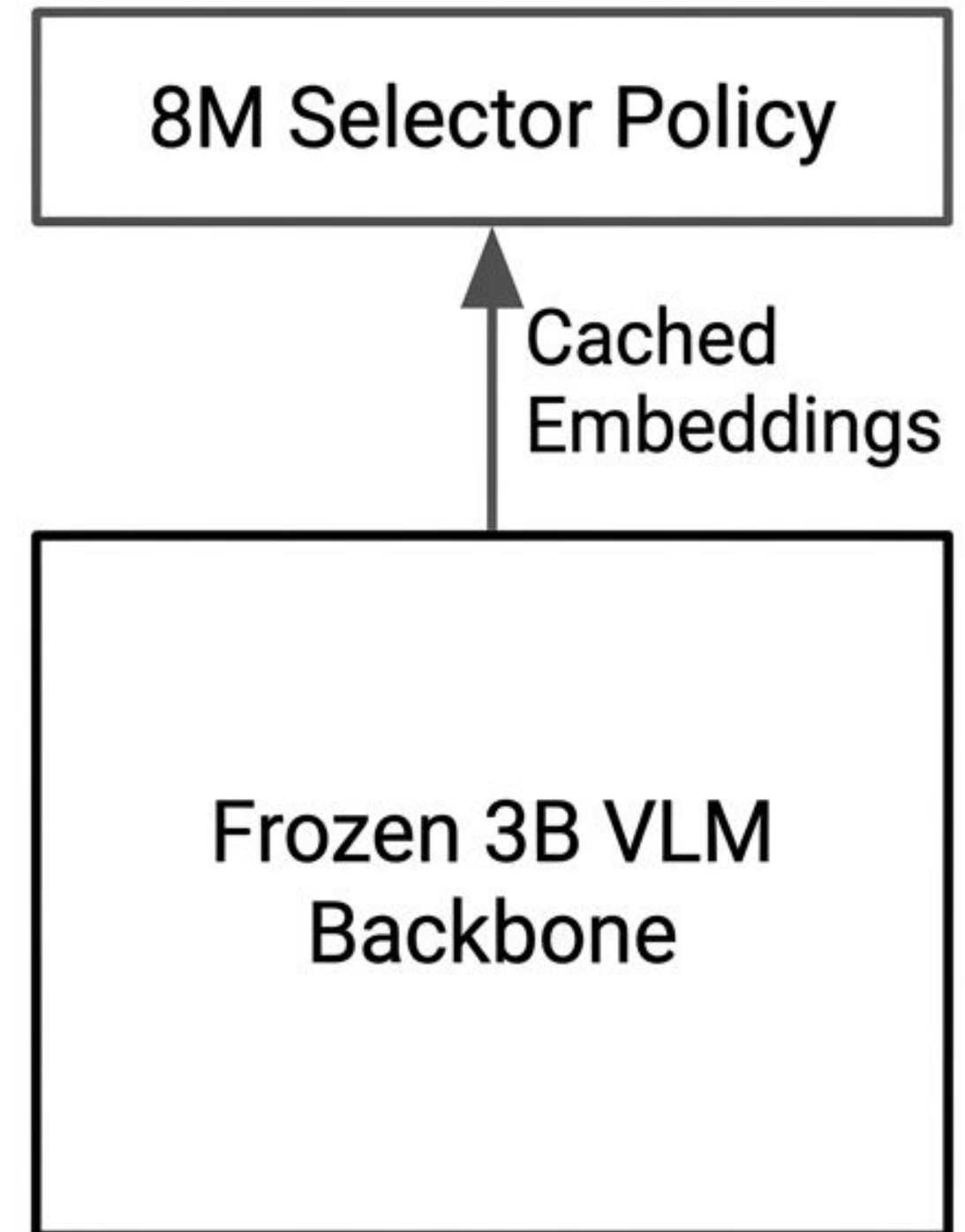
Environment Framing and the 3-Way Decision Gate

- **Efficiency Constraint:** The VLM is called exactly once per episode at the terminal step to produce the final answer.
- **Reward Structure:** Terminal-only. +1 for a correct answer, penalized by $\lambda \times$ (fraction of frames kept).



System Architecture

- **Selector Model:** 2-Layer Transformer with 8M parameters
 - Input → query + embedded frame + selected frames + progress scalars
 - Output → 3 layer softmax action head
- **Backbone:** Frozen Qwen2.5-VL-3B-Instruct
 - Input → query + selected frames
 - Output → MCQ answer
- **Dataset:** Next-QA for training and evaluation



Training Progression

- **Warm start** to imitate uniform selection and provide the policy with a baseline
- **8000 episodes of training**, KL divergence to limit divergence from warm start
 - Prevents catastrophic forgetting
- Standard PPO-clip using Generalized Advantage Estimation (GAE)

```
graph LR; A[Phase 0: Warm-Start] --> B[Phase 1: PPO Training];
```

Phase 0: Warm-Start

Behavioral cloning to imitate
uniform frame selection.

Phase 1: PPO Training

8,000 episodes using sparse
reward feedback

Preliminary Results on NExT-QA

- We compare to uniform sampling with different subset sizes (16/32)
- Using frame cost 0.2 dramatically drops the average frames and accuracy

Method	Accuracy	Frames
Uniform-16	76.0%	15.7
Uniform-32	78.0%	27.4
PPO ep800 ($\lambda=0.1$)	74.0%	13.3
PPO ep8000 ($\lambda=0.2$)	58.5%	1.5

The Penalty Trade-off and Early Stopping

- The frame cost (λ) is a highly sensitive hyperparameter
- Running with $\lambda=0.1$ across 800 episodes approaches uniform-16 accuracy (76%) with 2-3 fewer frames
- Using $\lambda=0.2$ (8000 episodes) results in stopping with 0-1 frames over 85% of the time, which tanks accuracy



Next Steps

- Retrain with $\lambda = 0.05$ and 0.1 to observe efficiency/accuracy tradeoff
- Analyze what types of questions (temporal, causal, descriptive) benefits and hurts the most from early stoppage

Thank you!